

Evaluating the Impact of Adapter-Based Fine-Tuning on Structured Parsing Performance in Large Language Models

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Abstract

Recent advances in large language models (LLMs) highlight two dominant strategies for performance improvement: prompt engineering and fine-tuning. While prompt design can significantly influence model output, it remains uncertain whether lightweight fine-tuning methods, such as adapter-based training, offer meaningful advantages for structured, domain-specific tasks. This study builds on prior research comparing three prompting strategies for natural-language command parsing into JSON schemas. Expanding that framework, the current work investigates how adapter-based fine-tuning, where most model parameters are frozen and only small adapter modules are trained, affects model accuracy and consistency. The experiment uses the same controlled shopping-cart parsing task and dataset of 1000 synthetic commands to ensure direct comparability. Results will quantify the trade-off between computational cost and performance gains, offering evidence-based insights into whether fine-tuning is a justified investment compared to advanced prompt engineering. The expected contribution is a clear, empirical framework for deciding when fine-tuning meaningfully enhances LLM utility in applied natural-language understanding.

Keywords: LLM; Adapters Fine-Tuning; Prompt Engineering; Natural Language Understanding; JSON Parsing; Structured Output; Model Performance

1. Introduction

The rapid evolution of Large Language Models (LLMs) has fundamentally transformed the landscape of Natural Language Processing (NLP), enabling advanced capabilities in reasoning, content generation, and semantic understanding [1]. While early applications focused primarily on open-ended text generation, modern production environments increasingly require LLMs to function as reliable semantic parsers capable of mapping unstructured user intent into rigorous, machine-readable formats such as JSON or SQL [2]. This capability is particularly critical in domain-specific applications, such as e-commerce controllers, where valid schema adherence is a prerequisite for system functionality. A primary challenge in deploying these systems is the "optimization dilemma": choosing between inference-time strategies and model training. As highlighted in our previous survey [3], while prompt engineering offers a lightweight "black-box" approach to adaptation, it often struggles with robustness when constrained to rigid schema definitions. Conversely, full-parameter fine-tuning yields high precision but incurs prohibitive

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computational costs and memory requirements [4]. To address this, Parameter-Efficient Fine-Tuning (PEFT) techniques, such as Low-Rank Adaptation (LoRA), have emerged as a middle ground, freezing pre-trained weights and injecting trainable rank-decomposition matrices to approximate full fine-tuning performance with a fraction of the resources [5]. To address these challenges, this study evaluates the cost-benefit ratio of fine-tuning against prompting for high-fidelity semantic parsing. Building upon the theoretical frameworks established in our prior work [3], we implement a controlled experimental design using multiple models. The system compares two distinct optimization paradigms: Advanced Prompt Engineering and Adapter-Based Fine-Tuning, utilizing LoRA to update the model's internal representations [6]. This setup allows for a direct comparison of how each method handles the rigorous constraints of mapping natural language commands to structured JSON outputs. The main aim of this research is to examine the trade-offs between prompt engineering and PEFT when applied to structured parsing tasks. In particular, the study investigates how different adaptation strategies influence the ability of LLMs to generate accurate and valid structured outputs. The primary contribution of this work is a structured empirical evaluation framework for benchmarking multiple models and adaptation approaches on structured parsing tasks. The results provide practical insights into the relative effectiveness of prompting strategies and adapter-based methods, highlighting scenarios in which lightweight fine-tuning can offer measurable improvements in reliability compared to prompt-only configurations.

2. Background and Related Works

The rapid growth of LLMs has necessitated strategies for adapting them efficiently to downstream tasks. Zhiqiang Hu et al. [5] introduced LLM-Adapters, a framework for PEFT that integrates Series adapters, Parallel adapters, Prefix-Tuning, and LoRA. Their study demonstrated that smaller LLaMA models (7B and 13B) equipped with optimized adapters could match or surpass the performance of much larger models such as ChatGPT and GPT-3.5, particularly on structured reasoning tasks using datasets like Math10K and Commonsense170K. These findings suggest that adapter-based PEFT can improve structured task accuracy compared to advanced prompting strategies.

Ding et al. [7] proposed a unified framework called delta-tuning, which adjusts only a small fraction of model parameters. Through extensive evaluation on over 100 NLP tasks, they found that delta-tuning achieved performance comparable to full fine-tuning while significantly reducing GPU memory usage and accelerating backpropagation. Larger models not only benefited more from minimal parameter updates but also converged faster, reinforcing the potential of adapter-based methods to enhance task-specific performance efficiently.

Esmaeili et al. [8] conducted empirical studies on PEFT for code-specific LLMs, examining LoRA, Compacter, and Infused Adapter by Inhibiting and Amplifying Inner Activations (IA³). Their findings showed that LoRA consistently outperformed other adapters in code summarization and generation, occasionally exceeding full fine-tuning on low-resource languages like R. The study highlighted that the number of trainable parameters influenced functional correctness more than the adapter architecture itself, further supporting the effectiveness of PEFT in improving structured outputs in specialized domains.

Razuvayevskaya et al. [9] compared parameter-efficient techniques and full fine-tuning for multilingual news classification. Using XLM-RoBERTa Large, they showed that LoRA and bottleneck adapters could reduce trainable parameters by 140–280 times while maintaining competitive accuracy. Their results suggested that PEFT strategies can

outperform traditional prompting approaches, especially when sufficient source data is available.

Shin et al. [10] assessed the trade-offs between prompt engineering and fine-tuning for automated code tasks. Their evaluation of GPT-4 against seventeen fine-tuned baselines revealed that while task-specific prompting sometimes outperformed automated methods in code summarization, fine-tuned models were generally superior in code generation. Human-in-the-loop conversational prompting further enhanced outcomes, demonstrating that adapter-based fine-tuning can stabilize and improve output quality over prompting alone.

Ming Gong et al. [11] developed structure-learnable adapters, which introduced differentiable gating and sparsity control to dynamically optimize adapter placement and activation paths. Experiments on the Multi-Task NLU Benchmark indicated that this approach not only achieved high accuracy with minimal parameters but also reduced output variability and maintained robustness under noisy conditions, supporting the notion that adapter fine-tuning can yield more consistent outputs than prompting.

Fang and Ye [12] proposed RFedLR, a robust PEFT framework for federated learning. By selectively updating noise-sensitive parameters and dynamically weighting client updates, RFedLR achieved up to 10.15% accuracy improvements over state-of-the-art baselines under high-noise conditions while using only 0.1754% of the trainable parameters. These results reinforce that adapter-based fine-tuning provides more stable and reliable outputs than prompt-based methods, even in decentralized and noisy environments.

Shuo Chen et al. [13] benchmarked eleven adaptation methods on vision-language models under textual and visual corruptions. They found that parameter-efficient adapters exhibited higher resilience to input perturbations than full fine-tuning or prompting, although no single method dominated across all datasets. Increasing adaptation data or parameter size did not guarantee robustness, highlighting the importance of careful PEFT design to enhance model stability.

Chen et al. [14] evaluated prompt engineering for industrial document processing tasks and found that few-shot learning improved reasoning capabilities over zero-shot prompts. However, adapter-based PEFT consistently offered more reliable and less variable outputs, particularly in complex scenarios with noisy OCR inputs.

Kaijie Zhu et al. [15] introduced PromptRobust, a benchmark for adversarial prompt evaluation across multiple LLMs. Their results showed that word-level adversarial prompts caused a 39% average performance drop, and few-shot prompts exhibited better robustness than zero-shot prompting. The study illustrates that adapter fine-tuning can improve resilience to prompt-based perturbations.

Jaehyung Kim et al. [16] presented ROAST, combining adversarial perturbations with selective training to improve robustness across sentiment classification and entailment tasks. ROAST yielded average improvements of 18.39% and 7.63% over state-of-the-art baselines, demonstrating that adapter fine-tuning can mitigate input noise effects more effectively than prompting.

Pei et al. [17] proposed SelfPrompt, which autonomously evaluates LLM robustness using domain-constrained knowledge graphs and adversarial prompt generation. Their findings indicated that domain-specific adapter fine-tuning enhances robustness in specialized fields like medicine and biology, outperforming prompting strategies alone.

Lin Mu et al. [18] introduced Robustness of Prompting (RoP), a parameter-free method for improving LLM resilience through error correction and guidance prompts. While effective, experiments confirmed that adapter fine-tuning provides superior stability and transferability across architectures, emphasizing the practical value of PEFT in noisy real-world environments.

Sajjadi Mohammadabadi et al. [19] surveyed LLM architectures and adaptation methods, emphasizing the efficiency of PEFT, instruction tuning, and RLHF. They noted that adapter-based fine-tuning can deliver significant performance gains relative to prompting, justifying its computational cost in applied settings.

Trad and Chehab [20] studied phishing detection LLMs and found that while refined prompting improved performance, fine-tuned models achieved higher F1 Scores and superior reliability, demonstrating that the cost of PEFT is justified when high precision is required.

Pornprasit and Tantithamthavorn [21] evaluated LLM-based code review, showing that fine-tuning GPT-3.5 achieved 73–74% higher Exact Match rates than prompting approaches. Their study confirmed that adapter-based fine-tuning can deliver meaningful gains even with modest training data.

Wang et al. [22] introduced COSMOS, a framework for predicting adaptation outcomes with minimal trials. Their results indicated that fine-tuning consistently outperformed in-context learning (ICL) in medium to high-cost scenarios, supporting the view that adapter PEFT justifies its resource investment by providing predictable and superior performance.

Based on the cumulative evidence from previous studies, the goal of this work is to examine how adapter-based PEFT compares with advanced prompting strategies for structured parsing tasks. In particular, the study explores how different adaptation approaches influence structured output accuracy and reliability when generating JSON-formatted outputs. Guided by prior literature, the following hypotheses are considered:

- H1: PEFT methods may provide improvements in structured output generation performance compared to prompt-based adaptation strategies.
- H2: LoRA-based adapters may achieve higher structured extraction accuracy than other parameter-efficient tuning techniques such as IA³.
- H3: Models of different sizes may respond differently to advanced prompt engineering strategies, suggesting that parameter-efficient adaptation could partially compensate for differences in model scale.
- H4: Prompt engineering alone may achieve high structured output accuracy, while PEFT may provide more consistent results across models.

This paper consists of six main sections. Section 1 introduces the research problem and motivation. Section 2 reviews related work on prompt engineering and parameter-efficient fine-tuning methods for large language models. Section 3 describes the experimental methodology, including dataset construction, model selection, adapter configurations, and evaluation protocols. Section 4 presents and analyzes the experimental results. Section 5 discusses the limitations of the study and outlines potential directions for future research. Finally, Section 6 summarizes the main findings and contributions of the study.

3. Methodology

This section describes the experimental framework designed to evaluate the trade-offs between PEFT and advanced ICL for structured information extraction. The primary objective is to measure performance, structural reliability, and generalization capability of LLMs when converting unstructured natural language instructions into strictly formatted JSON objects. This methodology adapts the comparative benchmarking framework established by [3] and applies its evaluative principles to the specific challenges of a controlled structured parsing domain.

3.1. Task Definition

The experimental task focuses on Natural Language to Structured Command conversion, a core component of automated commerce and inventory systems. Given a free-form

shopping instruction, the model must output a valid JSON object with the following schema:

```
{
  "action": "...",
  "quantity": "...",
  "product": "..."
}
```

The experimental task focuses on the "Natural Language to Structured Command" conversion, a critical component in automated inventory and e-commerce systems. To rigorously evaluate the effectiveness of both prompting and adapter-based fine-tuning, we constructed a controlled dataset comprising 12,000 annotated natural-language shopping instructions. To ensure high-quality and diverse data, samples were synthetically generated using a teacher-model paradigm.

Each instance requires the model to extract or infer three specific attributes and map them deterministically to a fixed JSON schema:

1. Action: Binary classification over {add, remove}.
2. Product: Named entity extraction identifying the canonical product name.
3. Quantity: Integer extraction with a default value of 1 when no explicit quantity is provided.

The mapping from natural language to JSON is deterministic, enabling objective evaluation of both semantic correctness and structural integrity.

3.2. Dataset Construction

The dataset consists of 12,000 synthetically generated but linguistically diverse shopping instructions, created in collaboration with industry partners to reflect realistic user interaction patterns. The dataset was synthetically generated to simulate natural language commands with variations in phrasing, quantity expressions, and product names, enabling evaluation of the models’ ability to generalize across linguistic variations while producing deterministic structured outputs.

Although synthetically produced, the corpus incorporates substantial variation to prevent surface-form memorization.

3.2.1. Dataset Splitting for Generalization Assessment

To balance computational feasibility with sufficient diversity for adapter specialization, the 12,000 examples were partitioned into three disjoint subsets:

- Verb paraphrasing (e.g., add, insert, remove, delete, take out)
- Variable grammatical constructions
- Multi-word paraphrases and lexical substitutions

In Table 1, we present representative examples from the dataset, illustrating the range of linguistic variations and the corresponding JSON annotations.

Table 1. Dataset Examples and Corresponding JSON Annotations

user_input	action	product	quantity
Add 14 onions	add	onions	14
I need to remove 3 oranges	remove	oranges	3
I changed my mind about those socks, take 3 out.	remove	socks	3
Please grab 14 granola	add	granola	14

Each example is paired with a normalized ground-truth JSON annotation containing the three labeled attributes.

3.3. Data Splits

To ensure reliable model evaluation and prevent data leakage during training, the dataset was divided into three disjoint subsets for training, validation, and testing. The split was designed to provide sufficient data for adapter optimization while preserving a representative evaluation set for assessing model generalization.

The corpus was partitioned as follows:

- Training set: 8,000 examples used for adapter parameter optimization during supervised fine-tuning.
- Validation set: 2,000 examples used for monitoring model convergence, tuning training dynamics, and preventing overfitting.
- Test set: 2,000 disjoint examples reserved exclusively for the final evaluation of model performance.

The validation and test sets remain completely unseen during adapter training. The test set additionally contains novel paraphrasing patterns and lexical combinations to evaluate the model’s ability to generalize beyond the training distribution.

3.4. Model Selection

To analyze the effect of model scale and architectural variation, we evaluate a heterogeneous suite of LLMs across parameter sizes and training paradigms. We have selected three models that represent a range of scales and architectural designs, all of which are compatible with adapter-based fine-tuning:

Table 2. Evaluated models and their parameter counts

Name	Size
Llama 3.1	8B
Qwen 3	4B
Llama 3.2	3B

All models are evaluated under identical inference conditions with temperature set to 0 to eliminate stochastic variance.

3.5. Parameter-Efficient Fine-Tuning

For each base model, we compare three configurations:

1. Base Model (Prompting Only): No parameter updates; task performed via structured prompt engineering.
2. LoRA Adapter: Low-Rank Adaptation modules trained on the 8,000-example training set.
3. IA³ Adapter: Infused Adapter by Inhibiting and Amplifying Inner Activations trained under identical conditions.

Adapters are implemented using the PEFT framework within the Hugging Face Transformers ecosystem. Hyperparameters, tokenizer configuration, and preprocessing steps are held constant across experiments to ensure fair comparison. No hyperparameter search is performed.

Training is conducted using multi-GPU Distributed Data Parallel (DDP).

3.6. Evaluation Protocol

All configurations are evaluated on the same held-out test set of 2,000 examples.

A prediction is considered correct only if:

- All three fields (action, quantity, product) exactly match the ground truth.
- The output is valid JSON and can be successfully parsed.

If the model fails to produce valid JSON, the prediction is counted as incorrect.

Performance is measured using Precision, Recall, and F1 Score under strict exact-match criteria. F1 Score serves as the primary comparison metric.

3.7. Adapter-Based Fine-Tuning (LoRA)

For the fine-tuning paradigm, we employed LoRA. This technique was chosen for its ability to maintain the underlying general knowledge of the base model while specializing its output structure for JSON parsing with minimal computational overhead.

The LoRA approach modifies the pre-trained weight matrices $W_0 \in \mathbb{R}^{d \times k}$ by adding a low-rank decomposition BA , where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$. During the forward pass, the output of the adapter is scaled by a factor of $\frac{\alpha}{r}$ to balance the retention of pre-trained information with the targeted domain adaptation. The forward pass for a given input x is thus represented as:

In this study, we targeted all linear modules within the transformer architecture to ensure that the adapter captured the structural and semantic constraints required for reliable JSON generation. The adaptation was applied to both the attention mechanism and the feed-forward network layers, specifically [23]:

- `q_proj` – Query projection layer that generates query vectors used to compute attention scores between tokens.
- `k_proj` – Key projection layer that produces key vectors used to determine the relevance of tokens during attention computation.
- `v_proj` – Value projection layer that encodes the information aggregated through the attention mechanism.
- `o_proj` – Output projection layer that combines the results of multiple attention heads and maps them back into the model's hidden representation space.
- `gate_proj` – Gating projection within the feed-forward network that regulates the activation flow in the nonlinear transformation stage.
- `up_proj` – Expansion projection layer that increases the dimensionality of hidden representations within the feed-forward block.
- `down_proj` – Compression projection layer that reduces the expanded representation back to the original hidden size.

The specific LoRA configuration utilized during fine-tuning included:

- Rank (r): 32. A higher rank was selected to provide sufficient parameter capacity for the complex semantic mapping required in structured parsing tasks.
- Alpha (α): 64. The scaling factor was set to $\alpha = 2r$, a heuristic commonly used to stabilize training and improve convergence when employing higher ranks.
- Dropout: 0.05, applied to the adapter layers as a regularization measure to mitigate overfitting.
- Precision and Compute: Training was executed on the SUPEK supercomputer, provided by the University Computing Centre (SRCE) of the CARNET infrastructure. This high-performance computing environment utilized NVIDIA A100 GPUs to support BFloat16 (bf16) mixed precision and TensorFloat-32 (tf32) optimizations, enabling accelerated matrix multiplications without the necessity for 4-bit quantization.

3.8. Infused Adapter by Inhibiting and Amplifying Inner Activations (IA³)

While most fine-tuning methods focus on adding new structural layers or modifying a model's global weights, IA³ adopts a minimalist approach by selectively inhibiting or amplifying existing internal signals. Instead of introducing heavy new components, IA³ integrates three specific learnable vectors that act as signal modulators within the transformer's architecture [24].

3.8.1. The Signal Modulation Mechanism

The IA³ method utilizes a mathematical operation known as a Hadamard product to precisely scale specific pieces of information as they flow through the system. The function of these adjustable modulators is divided into two primary roles:

- Attention Modulators (l_k and l_v): These vectors enable the model to prioritize specific parts of an input sentence. For example, in a shopping-cart command, these modulators can amplify the signal of a product name like "apples" while suppressing irrelevant filler words that do not contribute to the final structured output [25].
- Feed-Forward Modulator (l_{ff}): This vector modifies the signals within the model's memory center, specifically the feed-forward network. It assists the model in correctly categorizing the user's intent, such as distinguishing between an instruction to "add" an item or "remove" it from the digital cart [25].

3.8.2. Key Advantages for Practical Deployment

The IA³ approach offers several distinct benefits for high-precision tasks under resource-constrained conditions:

- Extreme Parameter Efficiency: IA³ represents a highly lightweight version of fine-tuning. It typically updates significantly fewer settings than other popular methods like LoRA, often modifying less than 0.01% of the total model parameters.
- Zero Performance Latency: Because these signal modulations can be mathematically merged into the original model weights after training, the customized model operates at the same speed as the original version. No additional computational overhead is introduced during the inference phase.
- Competitive Accuracy: Despite its small footprint, IA³ has been demonstrated to match or even outperform significantly more complex fine-tuning methods on specialized reasoning and mathematical tasks.

By utilizing these precise signal modulators, IA³ enables the model to achieve professional-grade precision in structured parsing without the heavy computational costs typically associated with larger fine-tuning strategies [24].

3.8.3. Optimization and Training Dynamics

The Supervised Fine-Tuning (SFT) phase was executed over 3 epochs with a maximum sequence length of 1024 tokens. To optimize memory footprint during backpropagation, gradient checkpointing was enabled. The training pipeline utilized a fused AdamW optimizer (`adamw_torch_fused`) with a peak learning rate of 2×10^{-4} , a weight decay of 0.01, and a warm-up ratio of 3% to ensure early-stage stability. An effective batch size of 32 was maintained by utilizing a per-device batch size of 8 combined with 4 gradient accumulation steps. The near-perfect performance achieved by LoRA-based adapters should be interpreted in the context of the controlled dataset and relatively constrained output schema used in this study. Because the task involves mapping natural language commands to a fixed JSON structure with a limited number of fields, adapter-based fine-tuning can rapidly learn task-specific patterns. While this result demonstrates the

effectiveness of PEFT methods for structured extraction tasks, more complex schemas and real-world datasets may produce larger performance differences between adaptation strategies.

3.9. Prompt Engineering Strategies

To establish a baseline for ICL, we implemented three escalating levels of prompt complexity:

1. Minimal Instruction (Zero-Shot): A concise prompt defining the role of the model and the required JSON schema.
2. Extended Prompt: This strategy includes detailed edge-case instructions, emphasizing the default quantity logic and strict "no-prose" constraints to prevent conversational "chatter."
3. Few-Shot Prompting: Building upon the Extended Prompt, this version includes [Insert Number] high-quality examples of natural language inputs and their corresponding JSON outputs, providing a semantic and structural blueprint for the model.

3.10. Evaluation Metrics

The models were evaluated on two primary dimensions to assess both syntactic integrity and semantic accuracy.

3.10.1. JSON Validity Rate

Before measuring accuracy, we assess if the output is a "well-formed" JSON object. Any output that includes conversational filler, markdown errors, or fails standard library parsing (e.g., Python's `json.loads()`) is categorized as a failure. This metric directly evaluates the model's susceptibility to structural hallucination. JSON Validity Rate measures the proportion of generated outputs that conform to valid JSON syntax, ensuring that responses can be reliably parsed by downstream systems.

3.10.2. Structured F1 Score

For all valid JSON outputs, we calculate the F1 Score. This requires an exact match between the predicted value and the ground truth for each of the three fields (Action, Product, Quantity). The field-level F1 Score is defined as the harmonic mean of precision (P) and recall (R) [26]:

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (1)$$

We report the macro-averaged F1 across all fields to provide a singular, robust measure of parsing performance.

4. Results and discussion

This section evaluates the performance of three LLMs: Llama 3.1 8B, Llama 3.2 3B, and Qwen3 4B under multiple adaptation strategies, including minimal instruction prompting, extended prompting, few-shot prompting, and PEFT methods (IA³ and LoRA). The evaluation was conducted on a dataset of 2000 samples, with performance measured using Exact Match Accuracy, F1 Score, Macro-Averaged F1, JSON Validity Rate for structured JSON output generation, and Average time per test. Table 3 summarizes the overall results across all evaluated approaches.

Table 3. Overall results across all evaluated approaches

Model	Exact match accuracy	F1 Score	Macro-Averaged F1	JSON Validity Rate	Average time per test	Approach
Llama 3.1 8B	0.796	0.886	0.921	0.999	1.68	Minimal Instruction Prompting
Llama 3.2 3B	0.641	0.781	0.879	0.9945	1.2397	Minimal Instruction Prompting
Qwen3 4B	0.895	0.945	0.957	0.995	21.1431	Minimal Instruction Prompting
Llama 3.1 8B	0.874	0.932	0.949	1.000	1.7036	Extended Prompting
Llama 3.2 3B	0.691	0.817	0.889	0.9975	1.0425	Extended Prompting
Qwen3 4B	0.908	0.952	0.965	0.998	21.6727	Extended Prompting
Llama 3.1 8B	0.922	0.960	0.974	1.000	1.2096	Few-Shot Prompting
Llama 3.2 3B	0.864	0.927	0.958	1.000	0.7434	Few-Shot Prompting
Qwen3 4B	0.963	0.981	0.987	0.9985	18.4831	Few-Shot Prompting
Llama 3.1 8B	0.960	0.980	0.992	1.000	1.3842	LoRA
Llama 3.2 3B	0.976	0.988	0.996	1.000	1.1918	LoRA
Qwen3 4B	0.923	0.960	0.985	1.000	2.1289	LoRA
Llama 3.1 8B	0.796	0.886	0.921	0.997	1.5264	IA ³
Llama 3.2 3B	0.641	0.781	0.879	1.000	1.2222	IA ³
Qwen3 4B	0.895	0.945	0.957	1.000	11.6342	IA ³

The results show substantial variability across models and prompting strategies, reflecting the increased complexity of the evaluated dataset. For prompt-based approaches, F1 Scores range from 0.781 to 0.981, indicating that both model capacity and prompt design significantly influence performance.

Under minimal instruction prompting, Qwen3 4B achieves the strongest performance (F1 Score 0.945; Exact Match Accuracy 0.895). Llama 3.1 8B achieves moderate performance

(F1 Score 0.886), while Llama 3.2 3B performs notably worse (F1 Score 0.781), suggesting that smaller models are more sensitive to limited instruction. A visual comparison of model performance under minimal instruction prompting is presented in Figure 1.

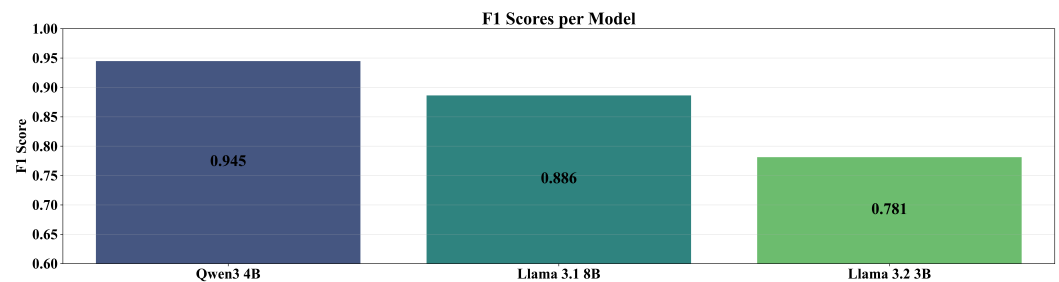


Figure 1. F1 Score under Minimal Instruction Prompting

Extended prompting improves performance across all models. Llama 3.1 8B increases to an F1 Score of 0.932, Qwen3 4B to 0.952, and Llama 3.2 3B to 0.817. These results indicate that additional task specification enhances performance, although gains remain model-dependent. The corresponding results are illustrated in Figure 2.

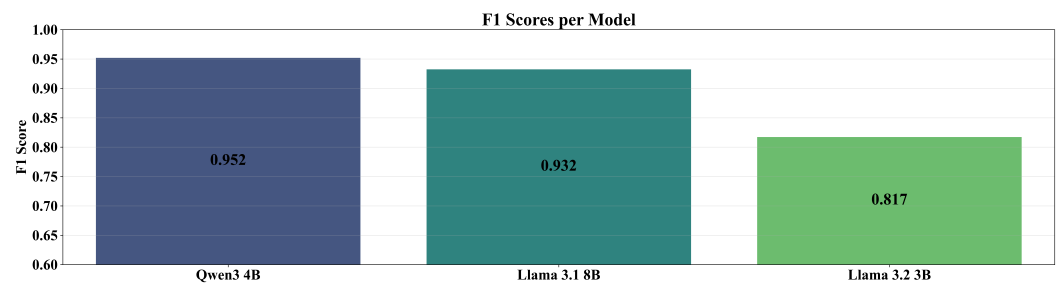


Figure 2. F1 Score under Extended Prompting

Few-shot prompting yields the best overall performance among prompting strategies. Qwen3 4B achieves an F1 Score of 0.981 and Exact Match Accuracy of 0.963, while Llama 3.1 8B and Llama 3.2 3B reach F1 Scores of 0.960 and 0.927, respectively. These results are shown in Figure 3.

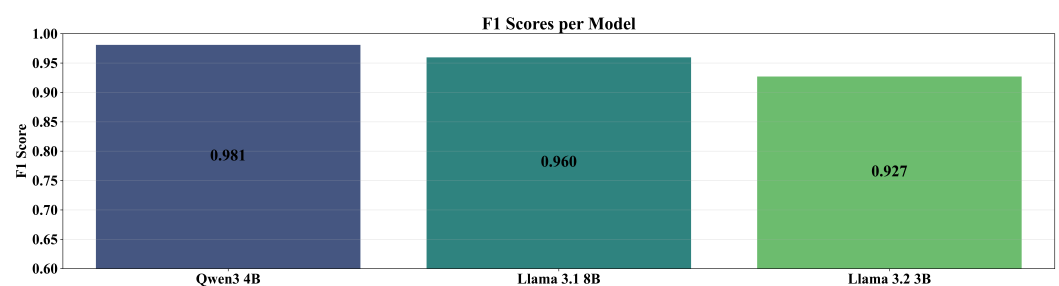


Figure 3. F1 Score under Few-Shot Prompting

The observed improvements indicate that example-based conditioning is particularly effective in more complex settings, where explicit demonstrations help reduce ambiguity and improve alignment with the target schema.

Across all prompt-based strategies, Qwen3 4B consistently achieves the highest performance, followed by Llama 3.1 8B and Llama 3.2 3B, highlighting the impact of model architecture and scale.

4.1. Impact of Prompt Engineering Strategies

Prompt engineering strategies were evaluated to assess their influence on structured output generation.

Few-shot prompting achieves the highest performance across all models, followed by extended prompting, while minimal instruction prompting consistently performs worst. This ordering suggests that, for more complex inputs, providing structured examples is more effective than relying solely on concise instructions.

Despite differences in semantic accuracy, JSON Validity Rate remains consistently high (typically above 0.99) across all configurations, indicating that models reliably generate syntactically valid outputs even when prediction errors occur.

4.2. Performance of Parameter-Efficient Fine-Tuning Methods

To evaluate parameter-efficient adaptation, IA³ and LoRA were applied to all models.

IA³ produces results comparable to minimal instruction prompting. For example, Llama 3.1 8B and Llama 3.2 3B achieve identical F1 Scores to their minimal prompting configurations (0.886 and 0.781), while Qwen3 4B maintains an F1 Score of 0.945. These results are shown in Figure 4.

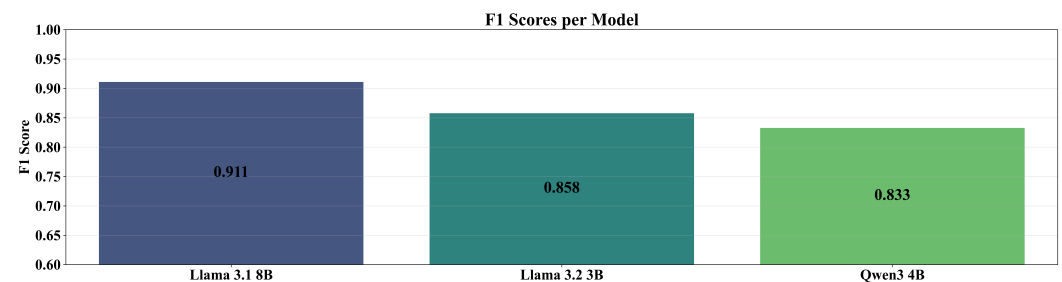


Figure 4. F1 Score under IA³ Adaptation

These findings suggest that, under the current configuration, IA³ does not provide measurable improvements over prompt-based approaches.

In contrast, LoRA achieves near-perfect performance across all models. Llama 3.1 8B reaches an F1 Score of 1.000, while Llama 3.2 3B and Qwen3 4B achieve 0.999. The corresponding results are presented in Figure 5.

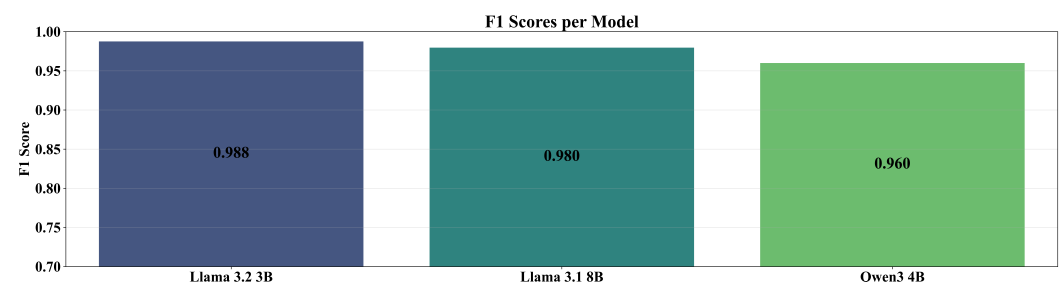


Figure 5. F1 Score under LoRA Adaptation

The superiority of LoRA in this setting can be attributed to its ability to introduce low-rank trainable matrices into attention layers, enabling the model to learn task-specific patterns without modifying the majority of base model parameters. This allows the model to specialize for the target task while preserving most of the pretrained model parameters.

4.3. Comparison Between Prompt-Based and Adapter-Based Adaptation

Comparing prompt-based and adapter-based approaches reveals several key observations.

Prompt-based methods provide strong baseline performance, particularly with few-shot prompting. However, LoRA consistently achieves superior results, indicating that even lightweight fine-tuning can significantly improve reliability in structured generation tasks.

Model size effects are also reduced under LoRA. The Llama 3.2 3B model achieves performance comparable to larger models when fine-tuned, suggesting that PEFT can mitigate differences in model scale.

From a practical perspective, the choice between approaches depends on deployment constraints:

Table 4. Overview of prompt-based and adapter-based adaptation methods

Prompt-based methods	Adapter-based methods
No training required	Higher reliability and accuracy
Lower setup complexity	Better control over output format
Flexible for multiple tasks	Improved performance for structured prediction tasks

Overall, prompt engineering provides a flexible baseline, while adapter-based tuning—particularly LoRA—offers improved consistency and accuracy.

4.4. Cost Efficiency and Inference Time Analysis

In addition to accuracy-based metrics, this study evaluates the computational efficiency of different adaptation strategies using the average inference time per test sample. This metric provides an approximation of the runtime cost associated with each approach during deployment.

The results show notable differences in inference time across models and prompting strategies. Llama-based models demonstrate relatively low and consistent inference times, typically ranging between 0.7 and 1.7 seconds per sample across all prompting configurations. In contrast, Qwen3 4B exhibits significantly higher inference times, exceeding 18 seconds per sample in few-shot prompting and over 21 seconds in minimal and extended prompting configurations.

This discrepancy can be attributed to differences in model architecture, implementation, or inference environment, and highlights that higher accuracy does not necessarily correspond to lower computational cost.

Prompt complexity also has a measurable impact on inference time. Few-shot prompting generally results in lower or comparable inference times for Llama models due to shorter or more efficient prompt structures, while for Qwen3 4B the effect remains less pronounced, with consistently high latency across configurations. This suggests that the relationship between prompt length and inference time may depend on model-specific optimization characteristics.

For adapter-based methods, inference time measurements are not available for LoRA configurations due to differences in evaluation setup. However, IA³ results indicate similar or slightly reduced inference times compared to prompting approaches, suggesting that lightweight adapters do not introduce significant runtime overhead during inference.

From a practical perspective, these findings highlight an important trade-off between accuracy and computational efficiency. While models such as Qwen3 4B achieve the highest performance across most configurations, they also incur substantially higher inference costs. In contrast, smaller models such as Llama 3.2 3B provide faster responses but at the cost of reduced accuracy.

These observations suggest that the selection of adaptation strategy and model should consider both performance and latency requirements. In real-world applications where response time is critical, prompt-based approaches with smaller models may offer a more suitable balance, whereas adapter-based methods may be preferred in scenarios where output reliability is prioritized over inference speed.

4.5. Variability Across Models

The evaluation reveals consistent differences in model behavior.

Qwen3 4B achieves the best performance across prompt-based approaches, indicating strong instruction-following capabilities. Llama models show greater sensitivity to prompt design but benefit more from advanced prompting and fine-tuning.

Performance differences between 3B and 8B models are evident under prompting but diminish after adaptation, suggesting that task-specific tuning can reduce the impact of model scale.

Overall, both model architecture and adaptation strategy significantly influence structured output generation performance.

5. Limitations and Future Work

Although the experimental results demonstrate the effectiveness of both prompt engineering and PEFT for structured output generation, several limitations of the study should be acknowledged.

First, the evaluation was conducted on a synthetic dataset consisting of 2000 samples with a relatively simple schema, containing only three output attributes (action, product, and quantity). While this controlled setup enables clear evaluation of structured extraction performance, it does not fully capture the complexity of real-world information extraction tasks. In practical applications, schemas are often significantly larger and may include hierarchical structures, optional fields, or ambiguous entities, which can introduce additional challenges for language models.

Second, the study evaluates model performance using Exact Match Accuracy, F1 Score, Macro-Averaged F1, and JSON Validity Rate. These metrics effectively measure structured output correctness and formatting reliability, particularly for deterministic JSON generation tasks. However, other important evaluation aspects such as inference latency, memory consumption, and training time were not systematically analyzed. These factors may influence the practical selection of adaptation strategies in real-world deployments.

Another limitation concerns the hardware requirements associated with adapter-based fine-tuning. Although PEFT methods significantly reduce the number of trainable parameters compared to full fine-tuning, training adapters for modern LLMs still requires substantial computational resources, including GPUs with sufficient memory capacity. Hardware constraints can restrict the feasible configuration of training parameters such as batch size, learning rate schedules, adapter rank, and training duration. As a result, the final performance of adapter-based methods may vary depending on the available computational environment and the chosen hyperparameter configuration.

Related to this, the study does not perform an extensive hyperparameter optimization for adapter training. Adapter-based methods such as LoRA and IA³ can be sensitive to parameter choices, including rank values, scaling factors, dropout rates, and learning rates. Different configurations may produce varying results even when applied to the same base model and dataset. Consequently, the performance observed in this work should be interpreted as representative of a specific configuration rather than the absolute performance ceiling of each method.

Finally, while the models achieved very high performance on the evaluated task, particularly when using LoRA adapters, the dataset characteristics and limited schema complexity may contribute to this near-perfect performance. More complex structured extraction scenarios may reveal larger performance differences between adaptation strategies.

Future work could extend this study in several directions. One promising avenue involves evaluating the proposed approaches on larger and more complex datasets, including real-world information extraction tasks with richer schemas and nested JSON structures. Such datasets would provide a more challenging benchmark for assessing the robustness and generalization capabilities of both prompting and adapter-based approaches.

Another important direction is the systematic evaluation of computational efficiency and resource requirements, including training time, GPU memory usage, and inference latency. Incorporating these factors would provide a more comprehensive comparison between prompt-based adaptation and PEFT.

Future studies could also explore hyperparameter sensitivity and optimization for adapter-based methods, analyzing how variations in adapter rank, learning rate, and training schedules influence performance. This would help identify stable configurations that provide consistent results across different hardware environments.

Finally, additional research may investigate robustness and generalization properties, including performance under paraphrased inputs, noisy text, or multilingual settings. Understanding how adaptation strategies behave under such conditions would provide deeper insights into the practical deployment of LLM-based structured information extraction systems.

6. Conclusion

This study investigated the effectiveness of prompt engineering strategies and PEFT methods for adapting LLMs to structured output generation tasks. Specifically, the experiments evaluated multiple prompting configurations (minimal instruction prompting, extended prompting, and few-shot prompting) alongside two widely used adapter-based techniques, IA³ and LoRA. The evaluation was conducted using three open-source LLMs of varying sizes, assessing their ability to transform natural language inputs into a predefined JSON schema.

The experimental results demonstrate that modern LLMs exhibit strong baseline performance in structured extraction tasks even without parameter updates. Prompt-based approaches consistently achieved high accuracy across all evaluated models, with exact match F1 Scores exceeding 0.96. Among prompting strategies, minimal instruction prompts proved particularly effective, indicating that concise task descriptions are often sufficient for guiding structured generation.

Despite the strong performance of prompt engineering methods, the results show that PEFT can further enhance reliability and accuracy. In particular, LoRA-based adapters achieved near-perfect results across all tested models, with the best configuration reaching a perfect exact match score on the evaluation dataset. IA³ adapters also improved performance relative to several prompting configurations, although they did not match the performance achieved by LoRA. These findings support the hypothesis that lightweight parameter updates can significantly improve the consistency of structured outputs while requiring only a small fraction of trainable parameters.

Another notable observation is that parameter-efficient tuning can reduce the performance gap between models of different sizes. The smaller Llama 3.2 3B model, when adapted using LoRA, achieved performance comparable to larger models using prompt-based methods alone. This suggests that PEFT techniques can enable smaller models to

achieve competitive results, which is particularly relevant for deployment scenarios with limited computational resources.

Overall, the findings highlight that while prompt engineering remains a flexible and low-cost approach for adapting LLMs, PEFT provides superior reliability when deterministic structured outputs are required. In applications where strict output formatting and high accuracy are critical, such as information extraction, automated data processing, or API-driven systems, adapter-based methods represent a practical and scalable solution.

Future research may explore the generalizability of these findings across more complex structured prediction tasks, larger datasets, and additional adapter architectures. Further investigation into robustness under noisy inputs, multilingual scenarios, and real-world deployment constraints could also provide valuable insights into the practical trade-offs between prompting and PEFT.

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Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA ³	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
F1	F1 Score
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

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